

Maryclare Griffin

Associate Professor
University of Massachusetts Amherst
Department of Mathematics and Statistics
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Amherst, MA 01003

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Employment and Appointments

Department of Mathematics and Statistics

University of Massachusetts Amherst

Associate Professor 2025-Present

Assistant Professor 2019-2025

Department of Mathematics and Statistics

University of Victoria

Affiliate Member of the Faculty of Graduate Studies 2026-2028

Center for Applied Mathematics

Cornell University

Postdoctoral Associate 2018-2019

Economics Research Center

University of Chicago

Research Assistant with Dr. James Heckman 2012-2013

Education

University of Washington, Seattle, WA

2013-2018

Ph.D., Statistics

Advisor: Peter Hoff, Thesis Title: “Model-Based Penalized Regression”

Duke University, Durham, NC

2016-2018

Visiting Ph.D. Student

University of Chicago, Chicago, IL

2008-2012

B.A., Economics with Honors, Statistics

Publications and Manuscripts

Submitted

MacArthur, I., Robacker, T., ..., Griffin, M., Reich, N. G., Johnson, K. E. “Collaborative estimation and evaluation of SARS-CoV-2 variant nowcasting in the United States.”

Submitted.

MacArthur, I., Robacker, T., Ray, E. L., Rogers, B. W., Reich, N. G., Griffin, M. “Comparison of probabilistic nowcasts and forecasts of SARS-CoV-2 variant proportions made by hierarchical multinomial linear regression models.”

Submitted.

Aprati, T. J., ..., Griffin, M., Kehl, K., Liu, D. “Clinico-genomic features predict distinct metastatic phenotypes in cutaneous melanoma.”

Submitted.

Kim D., Griffin M., Nolan D., Kitts J., Gile K. G. “Bayesian Resolution of Discrepant Self-Reported Network Ties.”

Submitted, available on request.

Griffin, M., “Review and Demonstration of a Mixture Representation for Simulation from Densities Involving Sums of Powers.”

Submitted.

Published

Griffin, M., Samorodnitsky, G., Matteson, D. S. “Likelihood Inference for Possibly Non-Stationary Processes via Adaptive Overdifferencing.”

Technometrics. 67(3):426-439.

Winn-Nuñez, E. T., Griffin, M., Crawford, L. (2024) “A Simple Approach for Local and Global Variable Importance in Nonlinear Regression Models.”

Computational Statistics and Data Analysis. 194:1-18

Griffin, M. (2024) “Improved Pathwise Coordinate Descent for Power Penalties.”

Journal of Computational and Graphical Statistics. 33(1):310-315.

Griffin, M., Hoff, P. D. (2024) “Structured Shrinkage Priors.”

Journal of Computational and Graphical Statistics. 33(1):1-14.

Selected for the **ASA Editors’ Choice Collection.**

Gelsinger, M., Griffin, M., Matteson, D. S., Guinness, J. (2023) “Log-Gaussian Cox Process Modeling of Large Spatial Lightning Data using Spectral and Laplace Approximations.”

Annals of Applied Statistics. 17(3): 2078-2094.

Zhang, W., Griffin, M., Matteson, D. S. (2023) “Modeling a Nonlinear Biophysical Trend Followed by Long-Memory Equilibrium with Unknown Change Point.”

Annals of Applied Statistics. 17(1):860-880.

Griffin, M., Hoff, P. D. (2020) “Testing Sparsity-Inducing Penalties.”

Journal of Computational and Graphical Statistics. 29(1):128-139.

Griffin, M., Hoff, P. D. (2019) “Lasso ANOVA Decompositions for Matrix and Tensor Data.”

Computational Statistics and Data Analysis. 137:181-194

Griffin, M., Gile K. J., Fredriksen-Goldsen K., Handcock M. S., Erosheva E. A. (2018) “A simulation-based framework for assessing the feasibility of respondent-driven sampling for estimating characteristics in populations of lesbian, gay and bisexual older adults.”

Annals of Applied Statistics. 12(4):2252-2278.

Holstein, C., Griffin, M., Hong J., Sampson P. (2015) “A Statistical Method for Determining and Comparing Limits of Detection of Bioassays.”

Analytical Chemistry. 87(19):9795-9801.

Software

Griffin, M. (2018) **gnorm**: Generalized Normal/Exponential Power Distribution for R, version 1.0.0.

Funding

National Science Foundation Division of Undergraduate Education 2022-2028
S-STEM Award (#2130262). Direct \$1,381,459, Indirect \$118,315.

National Science Foundation Division of Mathematical Sciences 2021-2024
Statistics Research Award (#2113079). Direct \$94,033, Indirect \$55,949.

Mutual Mentoring Team Grant. Direct \$5,502. 2020

National Science Foundation Graduate Research Fellowship 2013-2018

University of Washington, Blalock Fellowship 2013

Honors and Awards

College of Natural Sciences Inclusive Excellence Award	2025
College of Natural Sciences Faculty Peer Mentoring Award	2024
NSF Advance Fellow	2023-2024
Institute for Social Science Research (ISSR) Scholar	2021-2022
National Science Foundation Graduate Research Fellowship	2013-2018
Frontiers in Forecasting Best Poster Prize	2018
Women in Statistics and Data Science Conference Travel Award	2016
Survey Research Methods, Government Statistics, and Social Statistics Sections Student Paper Award	2015
University of Chicago, Goldberg Award in Economics	2012

Teaching

University of Massachusetts Amherst

Instructor for STAT 540/STAT 630: Statistical Learning	F2024, F2025
Instructor for STAT 535: Statistical Computing	F2023, S2025, F2026
Instructor for STAT 525: Regression Analysis	S2020, F2020, S2021, S2023
Instructor for STATISTC 697TS: Time Series Analysis and Applications	S2020, S2022

Cornell University

Instructor for STSCI4550/ILRST4550/ORIE5550: Applied Time Series Analysis	S2019
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University of Washington

Teaching Assistant for CSS&S564: Bayesian Statistics	S2016
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Postdoctoral and Student Advising

Dissertation Advisor

Isaac MacArthur	2026-Present
Thomas Robacker	2025-Present
Ning Duan	2024-Present
Yilin Zhou	2024-Present

Oral Exam Chair

Isaac MacArthur	Spring 2026
Thomas Robacker	Spring 2025
Ning Duan	Summer 2024
Yilin Zhu	Fall 2023

Honors Thesis Chair

Amel Hilmi	2025-2026
Maya Jacob	2024-2025
Gabrielle Walczak	2023-2024

Doctoral Dissertation Committee Member

Mike Wang, Department of Civil and Environmental Engineering	2026-Present
Yun Jiang, Department of Mathematics and Statistics,	2025-Present
Minsu Kim, Department of Biostatistics and Epidemiology	2025-2026
Gabriel Lewis, Department of Economics	2022-2024
Nutch Wattanachit, Department of Biostatistics and Epidemiology	2022-2023
Dongah Kim, Department of Mathematics and Statistics	2021-2022

Master's Thesis Committee Member

Puneet Velidi, Department of Mathematics and Statistics (University of Victoria)	2025-2026
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Oral Exam Committee Member

Cheng Wang, Department of Mathematics and Statistics	Spring 2025
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Minshen Xu, Department of Mathematics and Statistics	Spring 2023
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Yun Jiang, Department of Mathematics and Statistics	Spring 2023
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Undergraduate Research Mentoring

Camila Nunez Polanco (S-STEM REU), UMass Amherst	Summer 2025
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Natalie Pacheco (S-STEM REU), UMass Amherst	Summer 2024
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Trung Do (Department REU), UMass Amherst	Summer 2023
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Anh Pham Tran (Department REU), UMass Amherst	Summer 2023
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Vutjiya Senabunyarithi (Lee SIP REU), UMass Amherst	Summer 2023
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Gabrielle Walczak (NSF REU), UMass Amherst	Summer 2023
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Dayishaa Daga (Department REU), Mount Holyoke College	Summer 2021
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Thyra Tuttle, UMass Amherst	Spring 2021
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Assigned Visiting Assistant Professor Research Mentor	2020-2024,2025-2026
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Service**Associate Editor**

Journal of Computational and Graphical Statistics, Data Science in Science

Reviewer

Advances in Statistical Climatology, Meteorology and Oceanography; Annals of Applied Statistics; Applied Stochastic Models in Business and Industry; Bayesian Analysis; Biometrika; Biostatistics; Journal of Agricultural, Biological and Environmental Statistics; Journal of the American Statistical Association, Theory and Methods; Journal of Computational and Graphical Statistics; Journal of Econometrics; Journal of the Royal Statistical Society, Series A; NeuroImage; Observational Studies; PloS Computational Biology; The R Journal; Stat; Statistica Sinica; Statistics Surveys; Technometrics

NSF 2020-2021, 2023-2025

Department of Mathematics and Statistics, University of Massachusetts Amherst

Statistics Program Director	2026-2027
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Statistics Program Associate Director	2025-2026
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Undergraduate Affairs Committee	2023-2026
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Statistics & Probability Seminar Planning Committee	2019-2020, 2021-2022, 2023-2026
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Qualifying Exam Committee Member	Spring 2020-Spring 2022, Spring 2023-Fall 2024
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Major Advisor for ~10 Students/Year	Fall 2020-Spring 2022, Spring 2023-Spring 2025
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Anti-Racism Committee	2021-2025
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Onboarding Committee	2023-2024
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Statistics Faculty Search Committee (Elected)	Spring 2023, 2023-2024
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Bylaws Committee	2022-2023
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Statistics Graduate Admissions Committee	2021-2022
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Visiting Assistant Professor Faculty Search Committee (Elected)	2020-2021
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Five College Statistics Liaison	2020-2021
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Climate Committee	2019-2020
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University of Massachusetts Amherst

College of Natural Sciences Curriculum Committee	2025-Present
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Early Career Faculty Interview Committee for Associate Dean for Development	2021-2022
Computational Social Science Institute (CSSI) Steering Committee	2021-2022
Committee of Presidents of Statistical Societies (COPSS)	
COPSS Diversity, Equity, and Inclusion (DEI) Task Force	2020-2021
American Statistical Association	
Biometrics Section Continuing Education Chair	2024-2026
Biometrics Byar Award Committee Member	2024-2025
SLDS Section Student Paper Award Committee Member	2021-2024
ASA Task Force on Sexual Harassment and Assault	2018-2019
New England Statistics Society	
Student Competition Judge	2023-2024
University of Washington	
StatCom Member	2016-2018
Graduate Student Representative	2014-2015
Graduate and Professional Student Senate Representative	2013-2014

Presentations

Invited Talks

Not there yet: practical stopping criteria for MCMC with many chains.

SUNY Binghamton, Binghamton, NY; Aprile 2026.

Likelihood Inference for Possibly Nonstationary Processes via Adaptive Overdifferencing.

IMS International Conference on Statistics and Data Science.

Seville, Spain; December 2025.

Structured Shrinkage Priors.

Joint Statistical Meetings.

Nashville, TN; August 2025.

Forecasting and Nowcasting Variant Proportions with Genomic Data at the Regional Level.

Western North American Region of The International Biometric Society Conference.

Whistler, BC; June 2025.

Likelihood Inference for Possibly Nonstationary Processes via Adaptive Overdifferencing.

ASA/IMS Spring Research Conference.

New York, NY; June 2025.

Doing More with Less: Statistical Methods for High Dimensional and Correlated Regression Problems.

Brigham Young University, Provo UT; February 2025.

Bayesian generalized linear models for correlated data with fewer latent variables.

CMStatistics.

London, UK; December 2024.

Doing More with Less: Statistical Methods for High Dimensional and Correlated Regression Problems.

North Carolina State University, Raleigh NC; December 2024.

Doing More with Less: Statistical Methods for High Dimensional and Correlated Regression Problems.

Boston University, Boston MA; December 2024.

Doing More with Less: Statistical Methods for High Dimensional and Correlated Regression Problems.

Iowa State University, Ames IA; November 2024.

Bayesian Generalized Linear Models for Correlated Data with Fewer Latent Variables.

*Western North American Region of The International Biometric Society Conference.
Fort Collins, CO; June 2024.*

A Simple Approach for Local and Global Variable Importance in Nonlinear Regression Models.

New England Statistics Symposium, Storrs CT; June 2024.

Testing and estimation for sparsity-inducing power penalties.

Dartmouth College, Hanover NH; February 2024.

Testing and estimation for sparsity-inducing power penalties.

Rutgers University, New Brunswick, NJ; November 2023.

Log-Gaussian Cox Process Modeling of Large Spatial Lightning Data using Spectral and Laplace Approximations.

*The EnviBayes Workshop on Complex Environmental Data,
Colorado State University, Fort Collins, CO; September 2023.*

Log-Gaussian Cox Process Modeling of Large Spatial Lightning Data using Spectral and Laplace Approximations.

*Western North American Region of The International Biometric Society Conference.
Anchorage, AK; June 2023.*

Log-Gaussian Cox Process Modeling of Large Spatial Lightning Data using Spectral and Laplace Approximations.

New England Statistics Symposium, Boston, MA; June 2023.

Estimation of Possibly Non-Stationary Long Memory Processes via Adaptive Overdifferencing.

*Conference on Advances in Time Series Analysis with a Celebration of the
70th Birthday and Retirement of Professor Ruey Tsay, Chicago, IL; May 2023.*

An Introduction to Model-Based Penalized Regression.

*Department of Statistics,
Colby College, Waterville, ME; April 2023.*

Structured Shrinkage Priors.

*Department of Statistics,
University of Connecticut, Storrs, CT; April 2023.*

Likelihood Inference for Possibly Non-Stationary Processes via Adaptive Overdifferencing.

*Department of Mathematics and Statistics,
Washington University, Virtual; March 2022.*

Likelihood Inference for Possibly Non-Stationary Processes via Adaptive Overdifferencing.

*Booth School of Business, Econometrics and Statistics Colloquium,
University of Chicago, Chicago, IL; October 2021.*

Likelihood Inference for Possibly Non-Stationary Processes via Adaptive Overdifferencing.

New England Statistics Symposium, Providence, RI; October 2021.

Bayesian generalized linear models for correlated data with fewer latent variables.

CMStatistics, Virtual; December 2020.

Structured Shrinkage Priors.

*Department of Statistical Science,
University College London, Virtual; November 2020.*

Estimation of Possibly Non-Stationary Long Memory Processes via Adaptive Overdifferencing.

*Paul H. Chook Department of Information Systems and Statistics,
Baruch College, Virtual; October 2020.*

Estimation of Possibly Non-Stationary Long Memory Processes via Adaptive Overdifferencing.

*Department of Mathematics,
University of Maryland, Virtual; October 2020.*

A review of the computational aspects of penalized regression from a model-based perspective.

*Department of Mathematics and Statistics, Applied Mathematics and Computation
Seminar,
University of Massachusetts Amherst, Amherst, MA; February 2020.*

Estimation for possibly non-stationary long memory processes.

*Department of Biostatistics and Epidemiology,
University of Massachusetts Amherst, Amherst, MA; January 2020.*

Estimation for possibly non-stationary long memory processes.

*Computational Social Science Institute,
University of Massachusetts Amherst, Amherst, MA; November 2019.*

An Introduction to Model-Based Penalized Regression.

*Department of Mathematics and Statistics,
Amherst College, Amherst, MA; November 2019.*

Too Many Predictors, Too Few Responses.

*Research Bytes @ MassMutual,
Western Mass Statistics and Data Science, Amherst, MA; November 2019.*

Structured Shrinkage Priors.

*Department of Statistics,
George Mason University, Fairfax, VA; September 2019.*

Model-Based Penalized Regression.

*Department of Mathematics and Statistics,
University of Massachusetts Amherst, Amherst, MA; December 2018.*

Testing Sparsity-Inducing Penalties.

*Department of Biostatistics and Computational Biology Departmental Colloquia,
University of Rochester, Rochester, NY; October 2018.*

Beyond the Bayesian Lasso: A Review of Continuous Shrinkage Priors.

Joint Statistical Meetings, Vancouver, WA; August 2018.

Invited Panels

Wrapping Up and Moving On: Advice for the Final Stretch of Graduate School.

Women in Statistics and Data Science Conference, Virtual; October 2021.

Navigating the World of Data Science.

Voices of Data Science Conference, Virtual; February 2021.

Using Data to Inform the ASA's Policy on Sexual Misconduct.

International Conference on Health Policy Statistics, San Diego, CA; January 2020.

Contributed/Topic Contributed Talks

Forecasting and Nowcasting Variant Proportions with Genomic Data at the Regional Level.

*Joint Statistical Meetings,
Boston, MA; August 2026.*

Bayesian Generalized Linear Models for Correlated Data with Fewer Latent Variables.
Joint Statistical Meetings,
Portland, OR; August 2024.

Estimation of Possibly Non-Stationary Long Memory Processes via Adaptive Overdifferencing.
Joint Statistical Meetings,
Toronto, ON; August 2023.

Generalized Structured Shrinkage Priors Using Correlated Scales.
Joint Statistical Meetings, Virtual; August 2021.

Structured Shrinkage Priors.
Joint Statistical Meetings, Denver, CO; August 2019.

Long Memory Time Series Methods for the Analysis of Electronic Impedance Sensing Data.
Western North American Region of The International Biometric Society Conference.
Portland, OR; June 2019.

Structured Shrinkage Priors.
New England Statistics Symposium, Hartford, CT; May 2019.

Model-Based Testing of Sparsity Inducing Penalties.
Western North American Region of The International Biometric Society Conference.
Santa Fe, NM; June 2017.

Using Hierarchical Models to Understand P300-Wave-Based Brain-Computer Interface Performance Among Disabled Adults.
Joint Statistical Meetings, Chicago, IL; August 2016.

Assessing Feasibility of Respondent-Driven Sampling Using Pilot Data with an Application to Older Lesbian, Gay, and Bisexual Adults.
Joint Statistical Meetings, Seattle, WA; August 2015.

Invited Lightning Session

Log-Gaussian Cox Process Modeling of Large Spatial Lightning Data using Spectral and Laplace Approximations.
American Geophysical Union Fall Meeting, Virtual; December 2021.

Poster Presentations

Testing Sparsity-Inducing Penalties.
21st Meeting of New Researchers in Statistics and Probability.
Fort Collins, CO, August 2019.

Pathwise Coordinate Descent for Power Penalized Regression.
Non-convex Optimization and Deep Learning Workshop.
Boston, MA, January 2019.

Testing Sparsity-Inducing Penalties.
Bridging the Divide, Machine Learning in Medicine Symposium.
Ithaca, NY, September 2018.

Testing Sparsity-Inducing Penalties.
Cornell Day of Statistics.
Ithaca, NY, September 2018.

Testing Sparsity-Inducing Penalties.
Frontiers in Forecasting, Institute for Mathematics and Its Applications.
Minneapolis, MN, February 2018.

Sparse, Structured Matrix Estimation via ℓ_1 Penalization of ANOVA Decomposition.
Opening Workshop of the 2016-2017 Program on Optimization, SAMSI.
Research Triangle Park, NC, September 2016.

Workshops and Conferences

Topic Contributed Session Organizer, “Causal Inference with Interference: What Network and Spatial Statisticians Can Teach Each Other”

Joint Statistical Meetings, Boston, MA; August 2026.

Invited Session Organizer, “Novel Statistical Methods for Diverse Applications ”

New England Statistics Symposium, Storrs, CT; May 2026.

Topic Contributed Session Organizer, “Modern Statistical Methods and Applications that Enrich Society” and “Maintaining Privacy in Increasingly Public Societies”

Joint Statistical Meetings, Nashville, TN; August 2025.

Topic Contributed Session Organizer, “Innovative Adaptive Statistical Models for Time Series Data” and “Statistical Applications and Methods in Public Health: Highlights from Recent Mental, Reproductive, Maternal, and Adolescent Health Research with Policy Relevance”

Joint Statistical Meetings, Portland, OR; August 2024.

Topic Contributed Session Organizer, “Innovative Adaptive Statistical Models for Time Series Data” and “Statistical Applications and Methods in Public Health: Highlights from Recent Mental, Reproductive, Maternal, and Adolescent Health Research with Policy Relevance”

Joint Statistical Meetings, Portland, OR; August 2024.

Invited Session Organizer, “Finding and Plugging Leaks in the Pipeline to (Bio)Statistics Careers.”

Western North American Region of The International Biometric Society Conference,
Fort Collins, CO; June 2024.

Invited Session Organizer, “Beyond Independent and Identically Distributed: Models for Non-Standard Data”

New England Statistics Symposium, Storrs, CT; June 2024.

Invited Panel Organizer, “Women in the Statistics Workforce: Perspectives from Academia and Industry ”

Women in Statistics and Data Science Conference, Seattle, WA; October 2023.

Topic Contributed Session Organizer, “Bayesian Computation for Streaming, Shapes, and Selection”

Joint Statistical Meetings, Toronto, ON; August 2023.

Invited Session Organizer, “Advances in Scalable Regression Models for Complex Data.”

Western North American Region of The International Biometric Society Conference,
Anchorage, AK; June 2023.

Invited Session Organizer, “Spatial Statistics in a Changing World,” “Innovating in Causal Inference,” and “Cutting Edge Statistical Methods for Genetics and Genomics Data”

New England Statistics Symposium, Boston, MA; June 2023.

Invited Session Organizer and Chair, “Modern Solutions to Pressing Problems.”

CMStatistics, London, UK (Hybrid); December 2021.

Invited Session Organizer, “Innovations in exact and approximate time series analysis.”

New England Statistics Symposium, Providence RI (Hybrid); October 2021.

Invited Session Organizer, “Contemporary Mixed Model Methodology and Applications.”

Joint Statistical Meetings, Philadelphia, PA; August 2020.

Invited Session Organizer, “Modern and Practical Solutions to Difficult High Dimensional Regression Problems.”

Joint Statistical Meetings, Denver, CO; August 2019.

Invited Session Organizer, “Cutting Edge Methods for Modern Problems in Statistical Genetics and Genomics”

*Western North American Region of The International Biometric Society Conference,
Portland, OR; June 2019.*